Highlights

Cognitive flexibility (CF) is a fundamental ability, central to adaptive behaviour.

This meta-analysis shows that autistic people have greater CF difficulties.

These difficulties are present across the lifespan, from childhood into adulthood.

Certain tasks and outcome measures are more sensitive to detect differences.

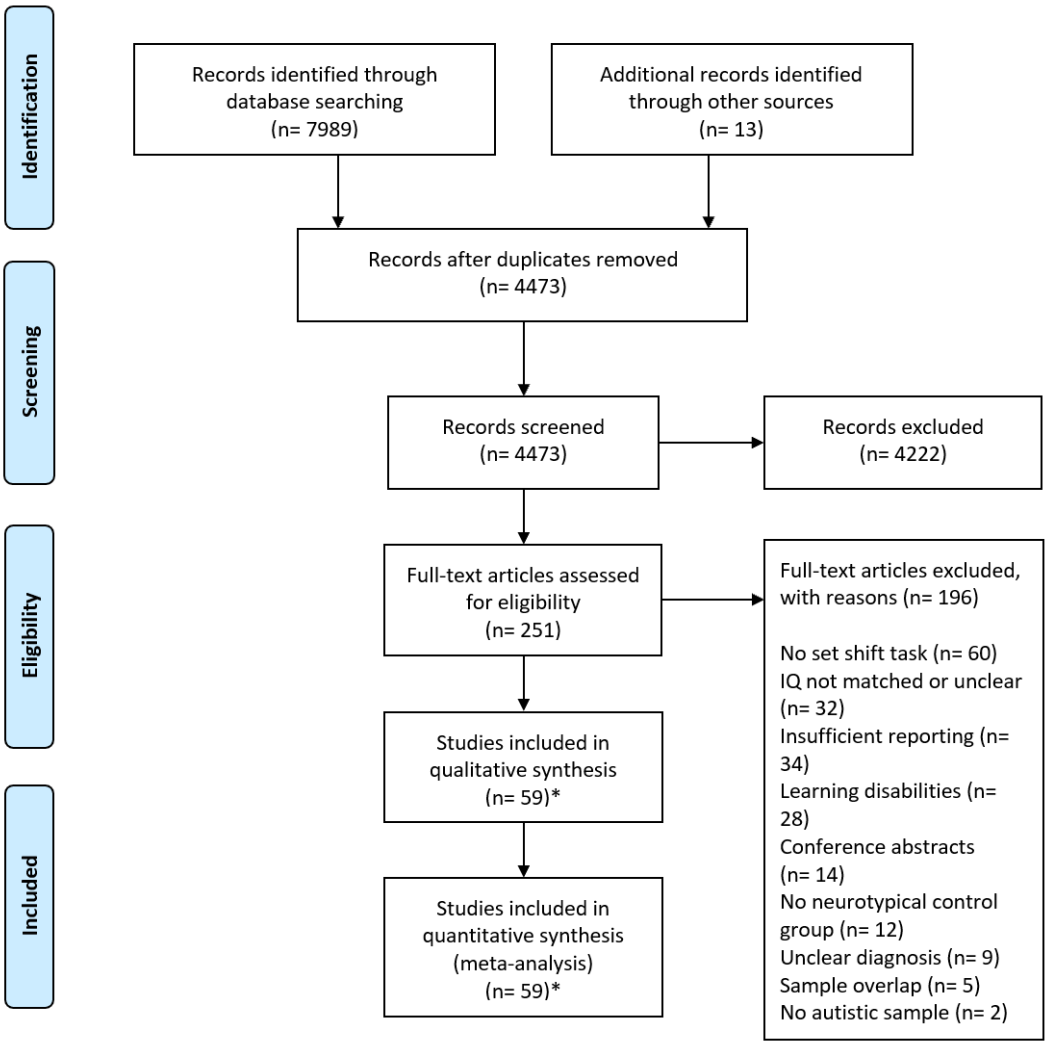
Heterogeneity of findings might suggest distinct profiles across the autism spectrum.

Abstract

Cognitive flexibility is a fundamental cognitive process that underlies adaptive behaviour in response to environmental change. Studies examining the profile of cognitive flexibility in autism spectrum disorder (ASD) have reported inconsistent findings. To address whether difficulties with cognitive flexibility are characteristic of autism, we conducted a random-effects meta-analysis and employed subgroup analyses and meta-regression to assess the impact of relevant moderator variables such as task, outcomes, and age. Fifty-nine studies were included and comprised of 2122 autistic individuals without intellectual disabilities and 2036 neurotypical controls, with an age range of 4 to 85 years. The results showed that autistic individuals have greater difficulties with cognitive flexibility, with an overall statistically significant small to moderate effect size. Subgroup analyses revealed a significant difference between task outcomes, with perseverative errors obtaining the largest effect size. In summary, the present meta-analysis highlights the existence of cognitive flexibility difficulties in autistic people, in the absence of learning disabilities, but also that this profile is characterised by substantial heterogeneity. Potential contributing factors are discussed.

Keywords: Autism Spectrum Disorder, Cognitive Flexibility, Set Shifting, Meta-Analysis

Figure 1



*Three studies (Crawley et al., 2020; Kado et al., 2020; Maister et al., 2013) had more than one experiment with no overlap in participants and were therefore included separately.

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Figure 3

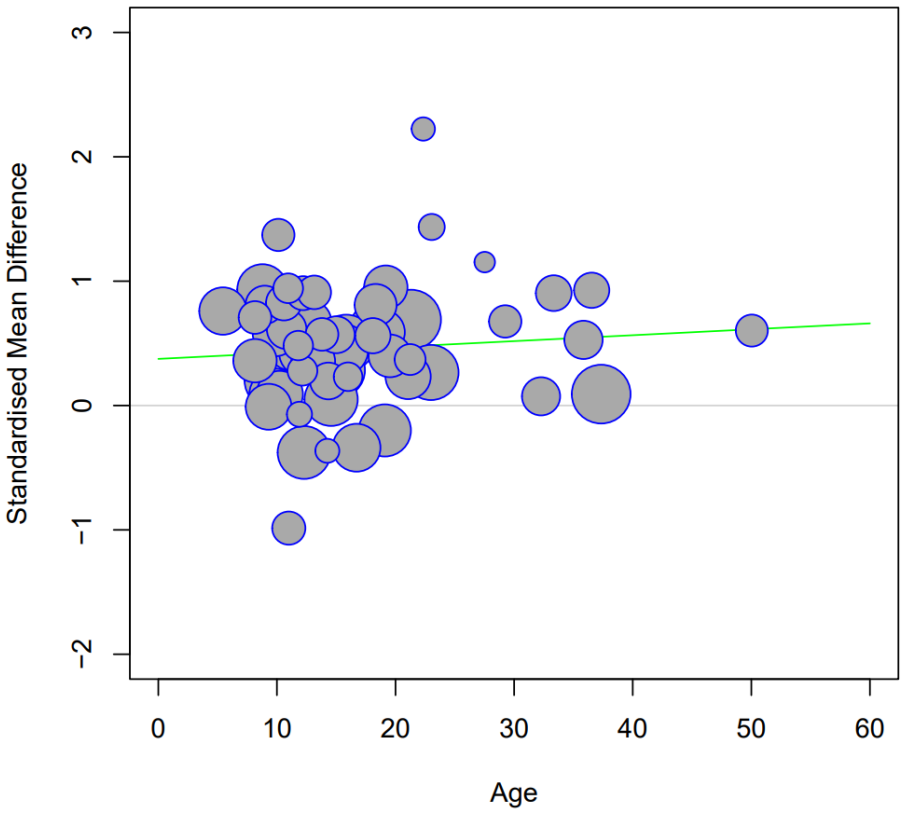


Figure 4

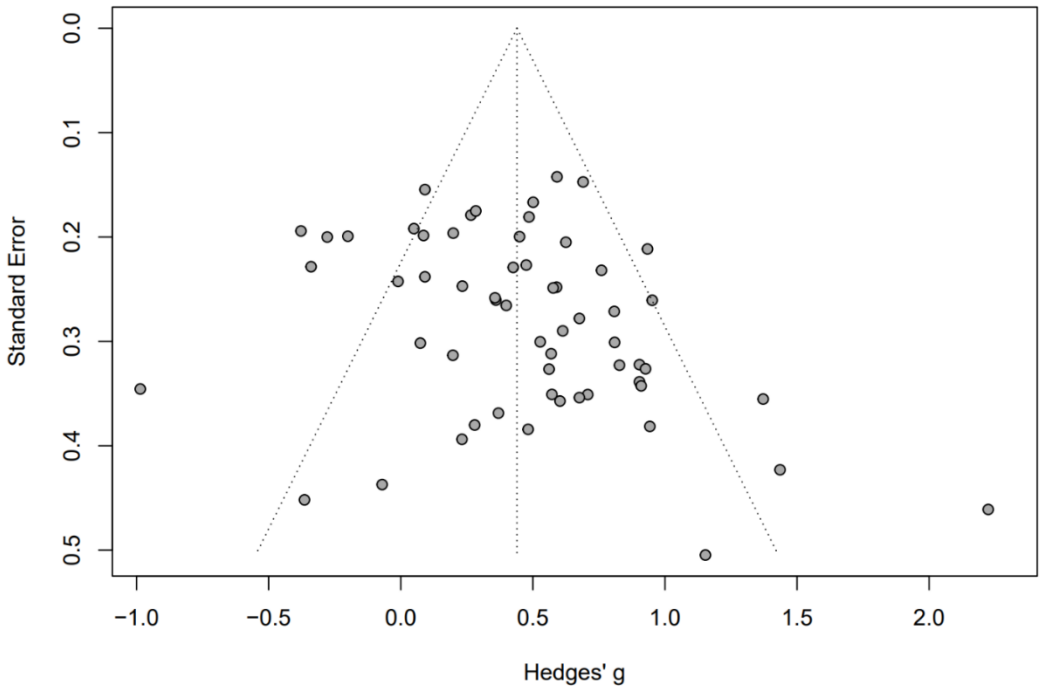


Figure 5

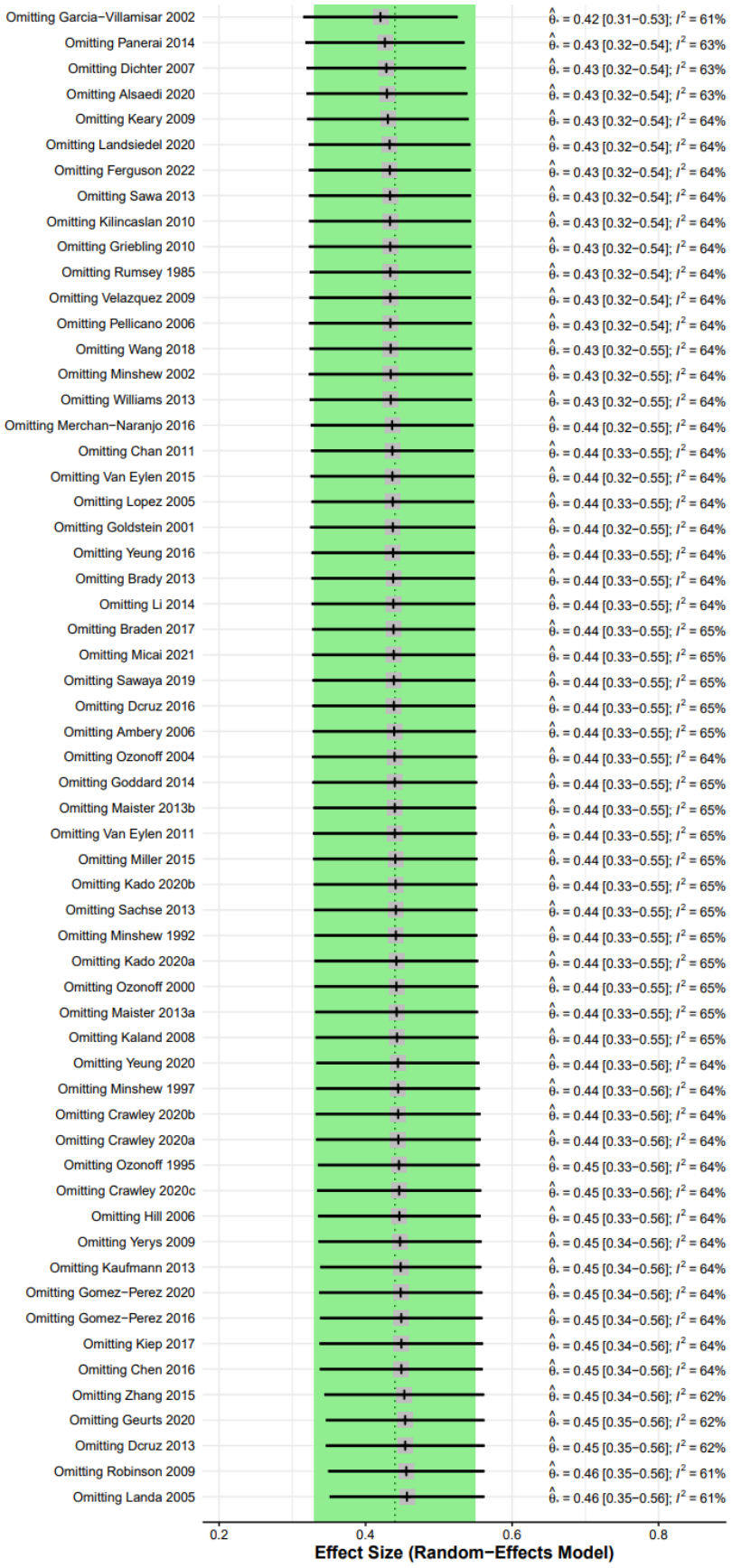
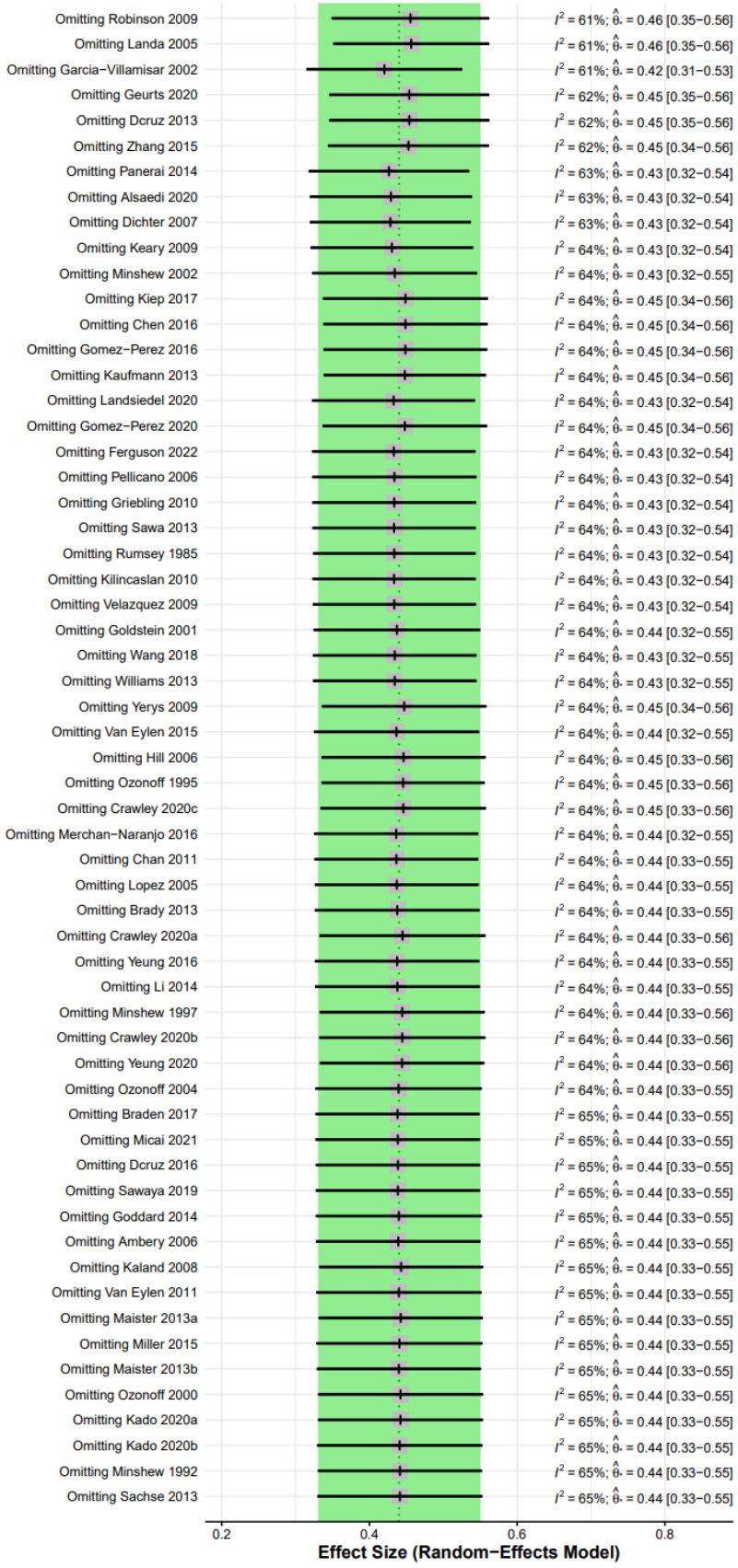
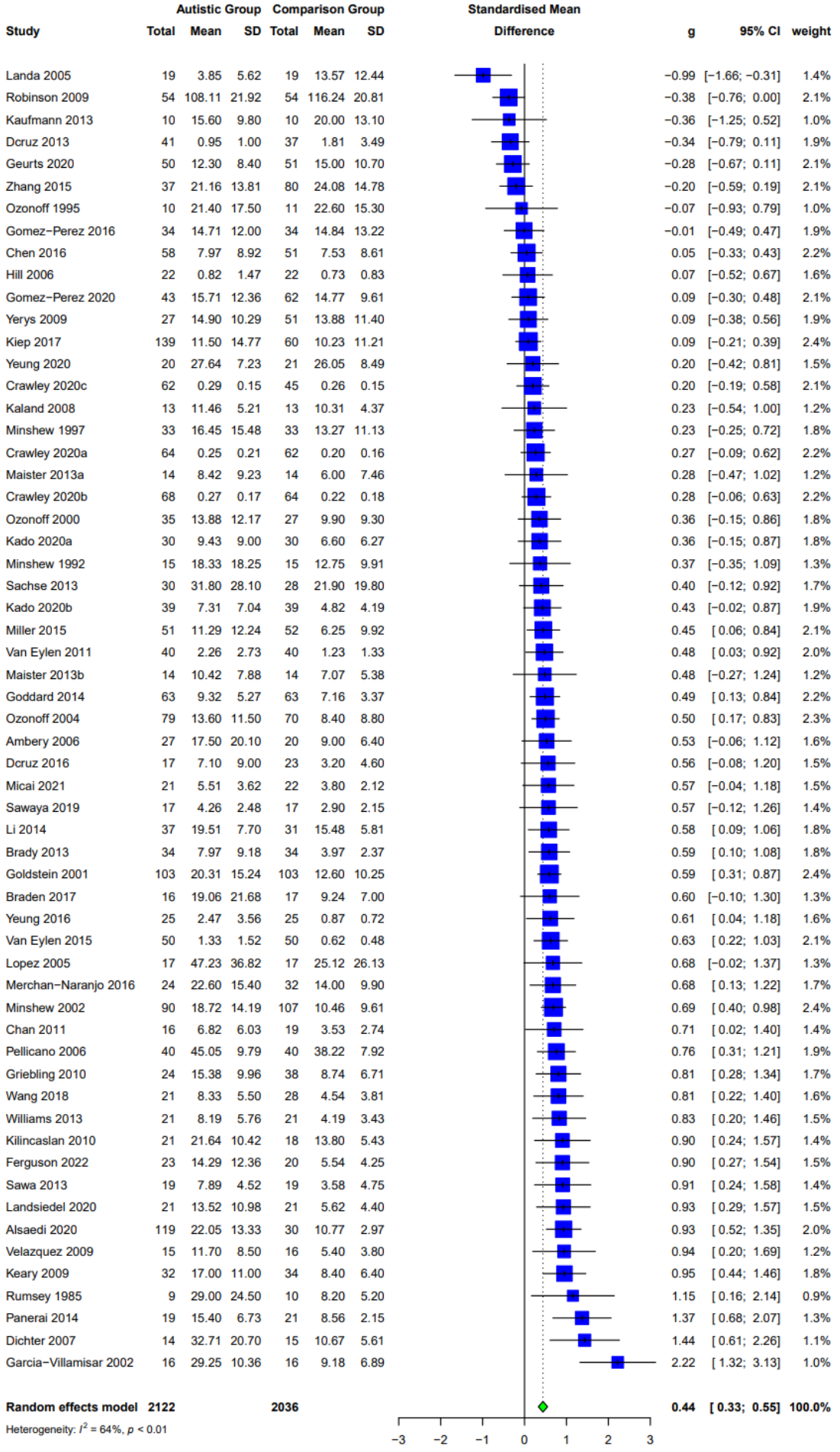

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Effect Size (Random−Effects Model)

Figure 6


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Figure 2



Tables 2-5 (Editable version)
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Table 2. Task Subgroup Results

	N	SMD	95% CI
WCST	22	0.58	0.37- 0.79
WCST-V	12	0.41	0.18- 0.64
IED	11	0.26	-0.01-0.53

SMD= Standardised Mean Difference (Hedges' g), CI= Confidence Interval, WCST= Wisconsin Card Sorting Test, WCST-V= Variations of WCST, IED= Intra-Extra Dimensional Set Shift Task,
Subgroup differences *p ≤ 0.05

Table 3. Outcome Subgroup Results

	N	SMD	95% CI
PE	43	0.48*	0.35-0.62
ESE	9	0.16*	-0.13-0.45

SMD= Standardised Mean Difference (Hedges' g), CI= Confidence Interval, PE= Perseverative Errors, ESE= Extra-Dimensional Shift Errors,
Subgroup differences *p ≤ 0.05

Table 4. Age Subgroup Results

	N	SMD	95% CI
Adults	22	0.54	0.34-0.73
Children	18	0.45	0.23-0.66
Adolescents	19	0.33	0.16-0.50

SMD= Standardised Mean Difference (Hedges' g), CI= Confidence Interval,
Subgroup differences *p ≤ 0.05

Table 5. Quality Assessment

Study	S1	S2	S3	S4	C1	E1	E2	E3	Total
Alsaedi 2020	✵	✵	✵	✵	✵✵	✵	✵	✵	9
Ambery 2006	-	-	-	-	-	✵	✵	✵	3
Braden 2017	✵	✵	✵	✵	✵✵	✵	✵	✵	9
Brady 2013	✵	-	-	-	✵	✵	✵	✵	5
Chan 2011	✵	-	✵	-	✵✵	✵	✵	-	6
Chen 2016	✵	-	✵	-	✵	✵	✵	✵	6
Crawley 2020a	✵	✵	✵	✵	✵✵	✵	✵	✵	9
Crawley 2020b	✵	✵	✵	✵	✵✵	✵	✵	✵	9
Crawley 2020c	✵	✵	✵	✵	✵✵	✵	✵	✵	9
D'Cruz 2016	-	✵	✵	✵	✵✵	✵	✵	✵	8
D'Cruz 2013	-	✵	✵	✵	✵✵	✵	✵	✵	8
Dichter 2007	-	-	✵	-	✵✵	✵	✵	✵	6
Ferguson 2022	✵	-	-	-	✵✵	✵	✵	-	5
Garcia-Villamisar 2002	-	-	✵	✵	✵✵	✵	✵	✵	7
Geurts 2020	✵	✵	✵	✵	✵	✵	✵	✵	8
Goddard 2014	✵	✵	✵	-	✵✵	✵	✵	✵	8
Goldstein 2001	-	-	-	✵	✵✵	✵	✵	✵	6
Gomez-Perez 2016	✵	✵	✵	-	✵✵	✵	✵	✵	8
Gomez-Perez 2020	✵	✵	✵	-	✵✵	✵	✵	✵	8
Griebling 2010	-	✵	✵	✵	✵✵	✵	✵	-	7
Hill 2006	✵	✵	✵	-	✵✵	✵	✵	✵	8
Kado 2020a	✵	-	-	-	✵	✵	✵	✵	5
Kado 2020b	✵	-	-	-	✵	✵	✵	✵	5
Kaland 2008	✵	-	✵	-	✵✵	✵	✵	✵	7
Kaufmann 2013	-	-	-	✵	✵✵	✵	✵	✵	6
Keary 2009	-	✵	✵	✵	✵✵	✵	✵	✵	8
Kiep 2017	✵	-	-	-	✵✵	✵	✵	✵	6
Kilincaslan 2010	-	-	✵	-	✵✵	✵	✵	✵	6
Landa 2005	-	-	✵	✵	✵✵	✵	✵	✵	7
Landsiedel 2020	✵	-	-	-	✵✵	✵	✵	-	5
Li 2014	✵	-	-	-	✵	✵	✵	-	4
Lopez 2005	✵	-	-	✵	✵✵	✵	✵	✵	7

Study	S1	S2	S3	S4	C1	E1	E2	E3	Total
Maister 2013a	✵	✵	✵	-	✵✵	✵	✵	✵	8
Maister 2013b	✵	✵	✵	-	✵✵	✵	✵	✵	8
Merchan-Naranjo 2016	✵	-	✵	✵	✵✵	✵	✵	✵	8
Micai 2021	✵	-	✵	-	✵✵	✵	✵	-	6
Miller 2015	✵	✵	✵	✵	✵✵	✵	✵	✵	9
Minshew 1992	✵	-	-	✵	✵✵	✵	✵	✵	7
Minshew 1997	✵	-	✵	✵	✵✵	✵	✵	✵	8
Minshew 2002	-	-	✵	✵	✵✵	✵	✵	✵	7
Ozonoff 1995	-	-	-	✵	✵✵	✵	✵	✵	6
Ozonoff 2004	-	-	-	-	✵✵	✵	✵	✵	5
Ozonoff 2000	✵	✵	-	✵	✵✵	✵	✵	✵	8
Panerai 2014	✵	✵	✵	-	✵✵	✵	✵	✵	8
Pellicano 2006	✵	✵	✵	✵	✵✵	✵	✵	✵	9
Robinson 2009	✵	✵	✵	-	✵✵	✵	✵	✵	8
Rumsey 1985	✵	-	-	-	-	✵	✵	✵	4
Sachse 2013	✵	-	✵	✵	✵✵	✵	✵	✵	8
Sawa 2013	✵	-	✵	✵	✵✵	✵	✵	✵	8
Sawaya 2019	✵	-	✵	-	-	✵	✵	✵	5
Van Eylen 2011	✵	-	✵	-	✵✵	✵	✵	✵	7
Van Eylen 2015	✵	-	✵	✵	✵✵	✵	✵	✵	8
Velazquez 2009	-	-	✵	-	-	✵	✵	✵	4
Wang 2018	-	-	✵	-	-	✵	✵	✵	4
Williams 2013	✵	-	-	✵	✵✵	✵	✵	✵	7
Yerys 2009	✵	-	✵	-	✵✵	✵	✵	✵	7
Yeung 2020	✵	✵	✵	✵	-	✵	✵	-	6
Yeung 2016	✵	-	✵	✵	✵✵	✵	✵	✵	8
Zhang 2015	✵	-	✵	✵	-	✵	✵	✵	6

Selection: S1 - Adequate case definition, S2 - Representativeness of cases, S3 - Selection of controls, S4 - Definition of controls

Comparability: C1 - Comparability of cases and controls on the basis of the design or analysis

Exposure: E1 - Ascertainment of exposure, E2 - Same method of ascertainment for cases and controls, E3 - Non-response rate

Table 1. Data Extracted from Included Studies

			Autistic Group (AG)		Comparison Group (CG)								
Source	**Age Range Mean (SD)**	**Task**	**Sample Size**	**Mean (SD)**	**Sample Size**	**Mean (SD)**	**Diagnostic Criteria**	**Diagnostic Tool**	**IQ Tool**	**Matched**	**FSIQ Mean**	**NVIQ/PIQ Mean**	**VIQ Mean**
Alsaedi 2020	6–12 AG 8.72 (1.96) CG 9.06 (1.42)	IED	119 M=95 F=24	TE 22.05 (13.33)	30 M=24 F=6	TE 10.77 (2.97)	DSM-IV-TR	--	RCPM	Age Sex NVIQ	--	AG 29.76 CG 29.80	--
Ambery 2006	19-67 AG 37.6 (14.6) CG 33.5 (12)	WCST	27 M=22 F=5	PE 17.5 (20.1)	20 M=16 F=4	PE 9.0 (6.4)	ICD-10	ADI ADOS	WAIS	Age Sex PIQ VIQ	--	AG 103.7 CG 109.4	AG 106.1 CG 107.05
Braden 2017	40–64 AG 50.1 (--) CG 50.0 (--)	WCST	16 M=16	PE 19.06 (21.68)	17 M=17	PE 9.24 (7)	DSM-IV DSM-5	ADOS	KBIT	Age FSIQ	AG 108.9 CG 110.2	--	--
Brady 2013	16–21 AG 18.86 (--) CG 18.90 (--)	IED	34 M= 76.5%	ESE 7.97 (9.18)	34	ESE 3.97 (2.37)	DSM-IV-TR	--	WASI	Age Sex FSIQ PIQ VIQ	AG 112.76 CG 110.44	AG 108.03 CG 109.03	AG 114.29 CG 109.32
Chan 2011	6-14 AG 7.98 (1.9) CG 8.30 (1.98)	MCST	16 M=16	PE 6.82 (6.03)	19 M=19	PE 3.53 (2.74)	DSM-IV	--	WISC	Age FSIQ VIQ	AG 89.50 CG 101.00	AG 98.90 CG 99.55	AG 78.40 CG 105.26
Chen 2016	13-18 AG 14.72 (1.53) CG 14.41 (1.42)	IED	58 M=57 F=1	ESE 7.97 (8.92)	51 M=50 F=1	ESE 7.53 (8.61)	DSM-IV	ADI-R	--	Age Sex FSIQ VIQ PIQ	AG 107.07 CG 109.92	AG 107.14 CG 110.80	AG 107.32 CG 107.53

Abbreviations: IED= Intra-extra dimensional set shift task, TE= Total errors, RCMP= Raven's coloured progressive matrices, WCST= Wisconsin card sorting test, PE= Perseverative errors, WAIS= Wechsler adult intelligence scale, KBIT= Kaufman brief intelligence test, WASI= Wechsler abbreviated scale of intelligence, ESE= Extra-dimensional shift errors, MCST= Modified card sorting test, WISC= Wechsler intelligence scale for children

Table 1. Data Extracted from Included Studies

			Autistic Group (AG)		Comparison Group (CG)								
Source	**Age Range Mean (SD)**	**Task**	**Sample Size**	**Mean (SD)**	**Sample Size**	**Mean (SD)**	**Diagnostic Criteria**	**Diagnostic Tool**	**IQ Tool**	**Matched**	**FSIQ Mean**	**NVIQ/PIQ Mean**	**VIQ Mean**
Crawley 2020a	-- AG 22.71 (3.60) CG 23.28 (3.10)	PRL	64 M=64%	PE 0.25 (0.21)	62 M=74%	PE 0.20 (0.16)	DSM-IV DSM-IV-TR DSM-5 ICD-10	ADOS ADI-R	Wechsler scales	Age Sex FSIQ	AG 109.73 CG 111.96	--	--
Crawley 2020b	-- AG 14.88 (1.73) CG 15.29 (1.76)	PRL	68 M=78%	PE 0.27 (0.17)	64 M=66%	PE 0.22 (0.18)	DSM-IV DSM-IV-TR DSM-5 ICD-10	ADOS ADI-R	Wechsler scales	Age Sex FSIQ	AG 106.76 CG 107.53	--	--
Crawley 2020c	-- AG 9.47 (1.46) CG 9.54 (1.57)	PRL	62 M=69%	PE 0.29 (0.15)	45 M=67%	PE 0.26 (0.15)	DSM-IV DSM-IV-TR DSM-5 ICD-10	ADOS ADI-R	Wechsler scales	Age Sex FSIQ	AG 109.11 CG 110.17	--	--
D'Cruz 2016	7-44 AG 17.4 (8.6) CG 18.6 (8.4)	2CRL	17 M=12 F=5	PE 7.1 (9.0)	23 M=18 F=5	PE 3.2 (4.6)	DSM-IV-TR	ADOS ADI-R	--	Age Sex FSIQ PIQ	AG 103.9 CG 110.9	AG 106.7 CG 107.5	AG 100.4 CG 113.0
D'Cruz 2013	8-44 AG 15.34 (7.75) CG 18.24 (8.12)	PRL	41 M=33 F=8	PE 0.95 (1.00)	37 M=31 F=6	PE 1.81 (3.49)	DSM-IV-TR	ADOS ADI-R	--	Age Sex FSIQ PIQ	AG 103.9 CG 108.7	AG 104.73 CG 107.59	AG 102.00 CG 109.00
Dichter 2007	-- AG 22.9 (5.2) CG 23.2 (5.7)	WCST-64	14 M=13 F=1	PE 32.71 (20.7)	15 M=14 F=1	PE 10.67 (5.61)	DSM-IV	ADI-R ADOS	WASI	Age Sex FSIQ PIQ VIQ	AG 105.0 CG 105.7	AG 104.1 CG 103.7	AG 105.1 CG 106.3

Abbreviations: PRL= Probabilistic reversal learning task (Crawley 2020: 80 trials with reversal midway + 80% trials accurately reinforced; D'Cruz 2013: reversal after 8 out of 10 consecutive correct responses + 80% trials accurately reinforced), PE= Perseverative errors, 2CRL= Two-choice reversal learning task, WCST= Wisconsin card sorting test, WCST-64= 64 cards total, WASI= Wechsler abbreviated scale of intelligence

Table 1. Data Extracted from Included Studies

			Autistic Group (AG)		Comparison Group (CG)								
Source	**Age Range Mean (SD)**	**Task**	**Sample Size**	**Mean (SD)**	**Sample Size**	**Mean (SD)**	**Diagnostic Criteria**	**Diagnostic Tool**	**IQ Tool**	**Matched**	**FSIQ Mean**	**NVIQ/PIQ Mean**	**VIQ Mean**
Ferguson 2022	-- AG 32.78 (11.14) CG 34.00 (11.16)	WCST-C	23	PE 14.29 (12.36)	20	PE 5.54 (4.25)	DSM-IV DSM-5 ICD-10	ADOS	WASI	Age Sex FSIQ PIQ VIQ	AG 104 CG 106	AG 103 CG 107	AG 104 CG 103
Garcia-Villamisar 2002	-- AG 23.5 (4.31) CG 21.19 (2.51)	WCST	16 M=8 F=8	PE 29.25 (10.36)	16 M=8 F=8	PE 9.18 (6.89)	DSM-IV	--	SPM	Age Sex NVIQ	--	AG 42.75 CG 43.69	--
Geurts 2020	60-85 AG 65.8 (5.6) CG 69.7 (5.6)	WCST-C	50 M=50	PE 12.3 (8.4)	51 M=51	PE 15 (10.7)	DSM-IV	--	WAIS	FSIQ	AG 110.7 CG 110.7	--	--
Goddard 2014	8-17 AG 12.5 (2.8) CG 12.1 (2.2)	WCST-64-C	63 M=51 F=12	PE 9.32 (5.27)	63 M=51 F=12	PE 7.16 (3.37)	DSM-IV-TR	--	WASI	Age Sex FSIQ	AG 103.6 CG 104.7	--	--
Goldstein 2001	-- AG 18.15 (10.14) CG 18.96 (10.1)	WCST	103 F= 13.6%	PE 20.31 (15.24)	103 F= 10.7%	PE 12.6 (10.25)	--	ADI ADOS	WISC WAIS	Age Sex FSIQ PIQ VIQ	AG 97.57 CG 99.12	AG 95.65 CG 98.6	AG 99.6 CG 99.94

Abbreviations: WCST= Wisconsin card sorting test, WCST-C= Computerised version, PE= Perseverative errors, WASI= Wechsler abbreviated scale of intelligence, SPM= Standard progressive matrices, WAIS= Wechsler adult intelligence scale, WCST-64-C= 64 cards total computerised version, WISC= Wechsler intelligence scale for children

Table 1. Data Extracted from Included Studies

			Autistic Group (AG)		Comparison Group (CG)								
Source	**Age Range Mean (SD)**	**Task**	**Sample Size**	**Mean (SD)**	**Sample Size**	**Mean (SD)**	**Diagnostic Criteria**	**Diagnostic Tool**	**IQ Tool**	**Matched**	**FSIQ Mean**	**NVIQ/PIQ Mean**	**VIQ Mean**
Gomez-Perez 2016	7-12 AG 9.35 (1.28) CG 9.26 (1.46)	WCST-64	34 M=30 F=4	PE 14.71 (12)	34 M=20 F=14	PE 14.84 (13.22)	DSM-IV DSM-5	--	WISC	Age FSIQ	AG 106.65 CG 108.12	--	--
Gomez-Perez 2020	7-13 AG 10.07 (1.65) CG 9.74 (1.56)	WCST-64	43 M=38 F=5	PE 15.71 (12.36)	62 M=33 F=29	PE 14.77 (9.61)	DSM-IV DSM-5	ADOS ADI-R	WISC	Age FSIQ	AG 94.16 CG 91.56	--	--
Griebling 2010	-- AG 17.9 (10) CG 18.6 (9)	WCST	24 M=22 F=2	PE 15.38 (9.96)	38 M=36 F=2	PE 8.74 (6.71)	--	ADOS ADI-R	WAIS WISC	Age Sex FSIQ	AG 104 CG 104	--	--
Hill 2006	16-64 AG 31.09 (13.14) CG 33.45 (14.54)	MCST	22 M= 16 F=6	PE 0.82 (1.47)	22 M=14 F=8	PE 0.73 (0.83)	DSM-IV	--	WAIS	Age FSIQ	AG 110.5 CG 107.91	--	--
Kado 2020a	5-9 AG 8.2 (1) CG 8.1 (1)	KWCST	30 M=22 F=8	PE 9.43 (9)	30 M=22 F=8	PE 6.6 (6.27)	DSM-IV-TR DSM-5	--	WISC	Age Sex FSIQ	AG 94 CG --	--	--
Kado 2020b	10-15 AG 12.2 (1.3) CG 12.2 (1.4)	KWCST	39 M=34 F=5	PE 7.31 (7.04)	39 M=34 F=5	PE 4.82 (4.19)	DSM-IV-TR DSM-5	--	WISC	Age Sex FSIQ	AG 96.9 CG --	--	--

Abbreviations: WCST= Wisconsin card sorting test, PE= Perseverative errors, WAIS= Wechsler adult intelligence scale, MCST= Modified card sorting test, WISC= Wechsler intelligence scale for children, WCST-64= 64 cards total, KWCST= Keio version WCST 48 cards+ sorting criterion change after 6 correct trials

Table 1. Data Extracted from Included Studies

			Autistic Group (AG)		Comparison Group (CG)								
Source	**Age Range Mean (SD)**	**Task**	**Sample Size**	**Mean (SD)**	**Sample Size**	**Mean (SD)**	**Diagnostic Criteria**	**Diagnostic Tool**	**IQ Tool**	**Matched**	**FSIQ Mean**	**NVIQ/PIQ Mean**	**VIQ Mean**
Kaland 2008	-- AG 16.4 (2.84) CG 15.6 (3.07)	WCST-C	13 M=13	PE 11.46 (5.21)	13 M=13	PE 10.31 (4.37)	ICD-10	ADI-R ADOS	WISC	Age FSIQ PIQ VIQ	AG 109 CG 109.62	AG 107 CG 106.92	AG 108.92 CG 110.15
Kaufmann 2013	-- AG 14.7 (5) CG 13.8 (5.3)	IED	10 M=8 F= 2	ESE 15.6 (9.8)	10 M=8 F=2	ESE 20 (13.1)	DSM-IV-TR	ADOS ADI-R	WISC WAIS	Age Sex FSIQ PIQ VIQ	AG 102.3 CG 109.5	AG 95.8 CG 106	AG 107.6 CG 114
Keary 2009	8-45 AG 19.8 (10.2) CG 18.6 (9.06)	WCST	32	PE 17 (11)	34	PE 8.4 (6.4)	--	ADI-R ADOS	WAIS WISC	Age FSIQ PIQ VIQ	AG 102.9 CG 104	AG 97.8 CG 102.6	AG 106.9 CG 104.7
Kiep 2017	19-60 AG 37.1 (9.8) CG 37.9 (11.1)	WCST-C	139 M=99 F=40	PE 11.5 (14.77)	60 M=35 F=25	PE 10.23 (11.21)	DSM-IV-TR	ADI-R	WAIS	Age FSIQ	AG 109.11 CG 110.71	--	--
Kilincaslan 2010	7-16 AG 12.44 (2.87) CG 11.96 (2.36)	WCST-C	21 M=18 F=3	PE 21.64 (10.42)	18 M=15 F=3	PE 13.8 (5.43)	DSM-IV	--	WISC	Age Sex FSIQ PIQ VIQ	AG 105.52 CG 107.27	AG 98.35 CG 107.44	AG 111.17 CG 106
Landa 2005	7-17 AG 11.01 (2.89) CG 11 (2.85)	IED	19	ESE 3.85 (5.62)	19	ESE 13.57 (12.44)	--	ADI-R ADOS	WISC WAIS	Age Sex FSIQ PIQ VIQ	AG 109.7 CG 113.4	AG 104.6 CG 108.5	AG 113.5 CG 115.6
Landsiedel 2020	-- AG 34.84 (11.42) CG 38.24 (13.19)	WCST-C	21	PE 13.52 (10.98)	21	PE 5.62 (4.4)	ICD-10 DSM-IV DSM-5	ADOS	WASI	Age Sex FSIQ PIQ VIQ	AG 104.32 CG 104.87	AG 102.92 CG 104.65	AG 105.2 CG 104.35

Abbreviations: IED= Intra-extra dimensional set shift task, WCST= Wisconsin card sorting test, PE= Perseverative errors, WAIS= Wechsler adult intelligence scale, WASI= Wechsler abbreviated scale of intelligence, ESE= Extra-dimensional shift errors, WISC= Wechsler intelligence scale for children, WCST-C= Computerised version

Table 1. Data Extracted from Included Studies

			Autistic Group (AG)		Comparison Group (CG)								
Source	**Age Range Mean (SD)**	**Task**	**Sample Size**	**Mean (SD)**	**Sample Size**	**Mean (SD)**	**Diagnostic Criteria**	**Diagnostic Tool**	**IQ Tool**	**Matched**	**FSIQ Mean**	**NVIQ/PIQ Mean**	**VIQ Mean**
Li 2014	6-12 AG 9.59 (2.29) CG 10.2 (1.53)	WCST-48-C	37	PE 19.51 (7.7)	31	PE 15.48 (5.81)	DSM-IV	--	RSPM	Age Sex NVIQ	--	AG 109.76 CG 113	--
Lopez 2005	18-45 AG 29.1 (8) CG 29.4 (11.4)	WCST	17 M=14 F=3	PE 47.23 (36.82)	17 M=11 F=6	PE 25.12 (26.13)	--	ADI-R ADOS	WAIS	Age Sex PIQ	AG 77 CG 89	AG 84.1 CG 87.6	AG 73 CG 92
Maister 2013a	11-13 AG 12.2 (0.6) CG 12.1 (0.2)	IED	14 M=14	ESE 8.42 (9.23)	14 M=13 F=1	ESE 6 (7.46)	--	ADI-R	RSPM	Age NVIQ	--	AG 43.9 CG 46.4	--
Maister 2013b	9-14 AG 11.8 (1.4) CG 11.8 (1.1)	IED	14 M=13 F= 1	ESE 10.42 (7.88)	14 M=11 F=3	ESE 7.07 (5.38)	--	ADI-R	RSPM	Age NVIQ	--	AG 41.6 CG 44.3	--
Merchan-Naranjo 2016	8-18 AG 12.8 (2.5) CG 12.9 (2.7)	WCST	24 M=23 F= 1	PE 22.6 (15.4)	32 M=30 F=2	PE 14 (9.9)	DSM-IV	ADOS	WAIS WISC	Age Sex FSIQ	AG 99.2 CG 106.81	--	--
Micai 2021	11-20 AG 15 (3) CG 15 (3)	WCST-C	21 M=17 F=5	PE 5.51 (3.62)	22 M=13 F=9	PE 3.80 (2.12)	--	ADOS	WAIS WISC	Age Sex FSIQ	AG 108 CG 115	--	--
Miller 2015	6-44 AG 15.1 (8.02) CG 15.9 (7.5)	PCET	51	RE 11.29 (12.24)	52	RE 6.25 (9.92)	DSM-IV	ADI-R ADOS	DAS WASI	Age Sex NVIQ	AG 100.1 CG 108.9	AG 101.1 CG 106.6	AG 100.1 CG 110.2
Minshew 1992	15-40 AG 21.13 (8.02) CG 21.33 (8.3)	WCST	15 M=15	PE 18.33 (18.25)	15	PE 12.75 (9.91)	DSM-III-R	ADI ADOS	WAIS	Age Sex FSIQ PIQ VIQ	AG 95.73 CG 96.47	AG 92.87 CG 93.27	AG 98.53 CG 99.07

Abbreviations: IED= Intra-extra dimensional set shift task, WCST= Wisconsin card sorting test, PE= Perseverative errors, WAIS= Wechsler adult intelligence scale, WASI= Wechsler abbreviated scale of intelligence, ESE= Extra-dimensional shift errors, WISC= Wechsler intelligence scale for children, WCST-48-C= 48 cards total computerised version, PCET= Penn conditional exclusion test, RE= Regressive errors, RSPM= Raven standard progressive matrices, DAS= Differential ability scales, WCST-C= Computerised version

Table 1. Data Extracted from Included Studies

			Autistic Group (AG)		Comparison Group (CG)								
Source	**Age Range Mean (SD)**	**Task**	**Sample Size**	**Mean (SD)**	**Sample Size**	**Mean (SD)**	**Diagnostic Criteria**	**Diagnostic Tool**	**IQ Tool**	**Matched**	**FSIQ Mean**	**NVIQ/PIQ Mean**	**VIQ Mean**
Minshew 1997	12-40 AG 20.91 (9.69) CG 21.21 (9.99)	WCST	33 M=29 F=4	PE 16.45 (15.48)	33 M=29 F=4	PE 13.27 (11.13)	--	ADI ADOS	WAIS	Age Sex FSIQ	AG 100.09 CG 100.48	AG 97.45 CG 99.09	AG 102.48 CG 101.30
Minshew 2002	-- AG 21.41 (9.68) CG 21.23 (9.81)	WCST	90	PE 18.72 (14.19)	107	PE 10.46 (9.61)	--	ADI-R ADOS	WAIS	Age Sex FSIQ PIQ VIQ	AG 97.95 CG 100.9	AG 95.51 CG 99.95	AG 100.11 CG 101.50
Ozonoff 1995	8-15 AG 11.9 (2.7) CG 11.9 (1.3)	WCST-C	10 M=9 F=1	PE 21.4 (17.5)	11 M=8 F=3	PE 22.6 (15.3)	DSM-IV	--	WISC	Age FSIQ PIQ VIQ	AG 98.1 CG 99.1	AG 101 CG 98.3	AG 95.8 CG 100
Ozonoff 2004	6-47 AG 15.7 (8.7) CG 16 (7.6)	IED	79 M=91%	ESE 13.6 (11.5)	70 M=83%	ESE 8.4 (8.8)	--	ADI-R ADOS	WISC WAIS	Age Sex FSIQ	AG 106.3 CG 106	AG 106 CG 105	AG 104.9 CG 106.1
Ozonoff 2000	6-20 AG 13.5 (4.05) CG 12.5 (3.2)	IED	35 M=31 F=4	ESE 13.88 (12.17)	27	ESE 9.9 (9.3)	DSM-IV	ADI-R ADOS	WISC	Age Sex FSIQ PIQ VIQ	AG 111.19 CG 111	AG 105.36 CG 110.6	AG 114.02 CG 109.9
Panerai 2014	-- AG 9.23 (3.31) CG 10.94 (2.87)	WCST	19 M=15 F=4	PE 15.4 (6.73)	21 M=14 F=7	PE 8.56 (2.15)	DSM-IV-TR	--	RCPM	Age Sex NVIQ	--	AG 23.58 CG 23.38	--
Pellicano 2006	4-7 AG 5.5 (0.9) CG 5.4 (0.9)	SST	40 M=35 F=5	TNTC 45.05 (9.79)	40 M=31 F=9	TNTC 38.22 (7.92)	DSM-IV	ADI-R	PPVT LIPS	Age Sex NVIQ VIQ	--	AG 113.58 CG 112.52	AG 101.15 CG 103.25

Abbreviations: IED= Intra-extra dimensional set shift task, RCPM= Raven's coloured progressive matrices, WCST= Wisconsin card sorting test, PE= Perseverative errors, WAIS= Wechsler adult intelligence scale, ESE= Extra-dimensional shift errors, WISC= Wechsler intelligence scale for children, WCST-C= Computerised version, SST= Set shifting task, TNTC= Total number of trials to criterion, PPVT= Peabody picture vocabulary test, LIPS= Leiter international performance scale

Table 1. Data Extracted from Included Studies

			Autistic Group (AG)		Comparison Group (CG)								
Source	**Age Range Mean (SD)**	**Task**	**Sample Size**	**Mean (SD)**	**Sample Size**	**Mean (SD)**	**Diagnostic Criteria**	**Diagnostic Tool**	**IQ Tool**	**Matched**	**FSIQ Mean**	**NVIQ/PIQ Mean**	**VIQ Mean**
Robinson 2009	-- AG 12.5 (2.7) CG 12.08 (2.3)	WCST-64-C	54 M=42 F= 12	PE 108.11 (21.92)	54 M=42 F=12	PE 116.24 (20.81)	DSM-IV	--	WASI	Age Sex FSIQ	AG 103.53 CG 104.80	--	--
Rumsey 1985	18-39 AG 27 (7) CG 28 (5)	WCST	9 M=9	PE 29 (24.5)	10 M=10	PE 8.2 (5.2)	DSM-III	--	WAIS	Age FSIQ PIQ	AG 104 CG 113	AG 104 CG 111	AG 103 CG 113
Sachse 2013	14-33 AG 19.2 (5.1) CG 19.9 (3.6)	IED	30 M=27 F=3	TE 31.8 (28.1)	28 M=24 F=4	TE 21.9 (19.8)	DSM-IV-TR	ADI-R ADOS	RSPM	Age Sex NVIQ	--	AG 105.3 CG 109.3	--
Sawa 2013	10-15 AG 13.2 (1.7) CG 13.1 (1.9)	KWCST	19 M=17 F=2	PE 7.89 (4.52)	19 M=17 F=2	PE 3.58 (4.75)	DSM-IV-TR	--	WISC	Age Sex FSIQ PIQ VIQ	AG 95.95 CG 97.32	AG 95.26 CG 97.84	AG 92.96 CG 97.21
Sawaya 2019	12-16 AG 14.6 (--) CG 13 (--)	CSG	17 M=16 F=1	RSE 4.26 (2.48)	17 M=11 F=6	RSE 2.9 (2.15)	--	ADI-R ADOS	WISC	Age NVIQ	--	AG 98 CG 95	AG 94 CG 104
Van Eylen 2011	8-18 AG 11.33 (2.18) CG 11.13 (2.22)	WCST-CTS	40 M=36 F=4	TE 2.26 (2.73)	40 M=36 F=4	TE 1.23 (1.33)	DSM-IV-TR	--	WISC WAIS	Age Sex FSIQ PIQ VIQ	AG 105.45 CG 106.76	AG 104.25 CG 103.88	AG 106.68 CG 109.65

Abbreviations: IED= Intra-extra dimensional set shift task, TE= Total errors, WCST= Wisconsin card sorting test, PE= Perseverative errors, WAIS= Wechsler adult intelligence scale, WASI= Wechsler abbreviated scale of intelligence, WISC= Wechsler intelligence scale for children, WCST-64-C= 64 cards total computerised version, KWCST= Keio version WCST 48 cards+ sorting criterion change after 6 correct trials, CSG= Computerised sequencing game, RSE= Rule shift errors, RSPM= Raven standard progressive matrices, WCST-CTS= Controlled task switching (shape & colour – sorting criterion changes randomly after 7, 8 or 9 correct trials)

Table 1. Data Extracted from Included Studies

			Autistic Group (AG)		Comparison Group (CG)								
Source	**Age Range Mean (SD)**	**Task**	**Sample Size**	**Mean (SD)**	**Sample Size**	**Mean (SD)**	**Diagnostic Criteria**	**Diagnostic Tool**	**IQ Tool**	**Matched**	**FSIQ Mean**	**NVIQ/PIQ Mean**	**VIQ Mean**
Van Eylen 2015	8-18 AG 12.21 (2.58) CG 12.48 (2.72)	WCST-CTS	50 M=30 F=20	PE 1.33 (1.52)	50 M=30 F=20	PE 0.62 (0.48)	DSM-IV-TR	--	WISC WAIS	Age Sex FSIQ PIQ	AG 104.32 CG 107.72	AG 104.32 CG 103.84	AG 104.32 CG 111.60
Velazquez 2009	7-16 AG 10.8 (3.4) CG 11.1 (2.6)	CST-C	15 M=14 F=1	PE 11.7 (8.5)	16 M=9 F=7	PE 5.4 (3.8)	DSM-IV	--	WASI	Age VIQ	AG 111.2 CG 123	--	AG 119.1 CG 119.9
Wang 2018	-- AG 9.05 (2.38) CG 8.92 (1.68)	MCST	21 M=20 F=1	PE 8.33 (5.5)	28 M=19 F=9	PE 4.54 (3.81)	DSM-IV	--	WISC	Age PIQ	AG 99.53 CG 110.14	AG 93.32 CG 101.36	AG 106.21 CG 116.04
Williams 2013	-- AG 10.6 (2.01) CG 10.59 (1.31)	MCST	21	PE 8.19 (5.76)	21	PE 4.19 (3.43)	DSM-IV ICD-10	--	WASI	Age PIQ VIQ	--	AG 110.19 CG 107.48	AG 103.57 CG 106.48
Yerys 2009	6-14 AG 10.19 (2) CG 10.26 (2.08)	IED	27	ESE 14.9 (10.29)	51	ESE 13.88 (11.4)	DSM-IV-TR	ADI-R ADOS	WASI WISC	Age Sex FSIQ	AG 111.95 CG 113.18	--	--
Yeung 2020	11-18 AG 14.43 (2.23) CG 14.27 (1.75)	PRL	20	TE 27.64 (7.23)	21	TE 26.05 (8.49)	DSM-5	ADI-R	WISC	Age Sex FSIQ	AG 104.18 CG 106.73	--	--
Yeung 2016	6-17 AG 10.09 (2.58) CG 11.55 (3.53)	WCST-CTS	25 M=19 F=6	PE 2.47 (3.56)	25 M=14 F=11	PE 0.87 (0.72)	DSM-IV	ADI-R	WISC	Age Sex FSIQ	AG 105.24 CG 110.48	--	--
Zhang 2015	-- AG 18.9 (3.64) CG 19.2 (2.96)	WCST	37 M=31 F=6	PE 21.16 (13.81)	80 M=67 F=13	PE 24.08 (14.78)	DSM-IV-TR	--	RSPM	Age Sex NVIQ	--	AG 103.2 CG 108.1	--

Abbreviations: IED= Intra-extra dimensional set shift task, TE= Total errors, WCST= Wisconsin card sorting test, PE= Perseverative errors, WISC= Wechsler intelligence scale for children, WAIS= Wechsler adult intelligence scale, WASI= Wechsler abbreviated scale of intelligence, ESE= Extra-dimensional shift errors, MCST= Modified card sorting test, WCST-CTS= Controlled task switching (shape & colour – sorting criterion changes randomly after 7, 8 or 9 correct trials), RSPM= Raven standard progressive matrices, CST-C= Card sorting task computerised, PRL= Probabilistic reversal learning task (Yeung 2020: reversals after 8 out of 10 consecutive correct responses + 80% trials accurately reinforced)

A Meta-Analysis of Cognitive Flexibility in Autism Spectrum Disorder

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Abstract

Cognitive flexibility is a fundamental cognitive process that underlies adaptive behaviour in response to environmental change. Studies examining the profile of cognitive flexibility in autism spectrum disorder (ASD) have reported inconsistent findings. To address whether difficulties with cognitive flexibility are characteristic of autism, we conducted a random-effects meta-analysis and employed subgroup analyses and meta-regression to assess the impact of relevant moderator variables such as task, outcomes, and age. Fifty-nine studies were included and comprised of 2122 autistic individuals without intellectual disabilities and 2036 neurotypical controls, with an age range of 4 to 85 years. The results showed that autistic individuals have greater difficulties with cognitive flexibility, with an overall statistically significant small to moderate effect size. Subgroup analyses revealed a significant difference between task outcomes, with perseverative errors obtaining the largest effect size. In summary, the present meta-analysis highlights the existence of cognitive flexibility difficulties in autistic people, in the absence of learning disabilities, but also that this profile is characterised by substantial heterogeneity. Potential contributing factors are discussed.

Keywords: Autism Spectrum Disorder, Cognitive Flexibility, Set Shifting, Meta-Analysis

1. Introduction

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition defined by difficulties in social communication and interaction, along with restricted and repetitive behaviours (DSM-5, American Psychiatric Association, 2013). Although a clear understanding of the causes of ASD remains elusive, several genetic and neurobiological factors have been identified (Lord et al., 2020). Several cognitive theories have also been proposed to underpin the ASD phenotype (e.g., Baron-Cohen et al., 1985; Frith, 1989). The executive dysfunction account of autism, originally proposed by Damasio and Maurer (1978), drew parallels between the rigid and perseverative behaviours of patients with frontal lobe lesions and those diagnosed with autism. Since then, extensive research over the last four decades has highlighted the existence of executive function (EF) difficulties in autistic people (e.g., Rumsey, 1985; Alsaedi et al., 2020) and neuroimaging studies have demonstrated structural and functional alterations in frontal regions (Catani et al., 2016; Ecker, 2017; Libero et al., 2015). Although no consensus exists, EF is often divided into the subdomains of working memory, planning, inhibition and cognitive flexibility, and evidence suggests that EF subdomains are separable, yet correlated (Miyake & Friedman, 2012).

Cognitive flexibility is a multifaceted construct involving the capacity to intentionally shift between different mental tasks or strategies and adjust responses according to changing environmental contingencies and is commonly assessed using set shifting paradigms (Dajani & Uddin, 2015). Set shifting entails using one set of rules and then changing to a different set to complete the same task, and it is often referred to as the most fundamental form of cognitive flexibility (Dajani & Uddin,

2015; Yerys et al., 2015). According to the hierarchy proposed by Bunge & Zelazo (2006), more complex forms of cognitive flexibility – such as switching between tasks – would depend on the integrity of set shifting abilities. In autistic people, difficulties with cognitive flexibility have been associated with increased social difficulties (Berger et al., 2003), increased restricted and repetitive behaviours (RRBs) (Faja & Darling, 2019; Miller et al., 2015) and co-occurring symptoms such as anxiety and low mood (Crawley et al., 2020; Ozsivadjian et al., 2021). Growing evidence also suggests a key role for cognitive flexibility in outcomes such as academic achievement (John et al., 2018), adaptive behaviour (Bertollo et al., 2020) and quality of life (de Vries & Geurts, 2015). Given that a substantial proportion of autistic adults without intellectual disabilities report lower rates of employment and independent living (Anderson et al., 2014; Frank et al., 2018), fewer relationships (Farley et al., 2009) and reduced quality of life (Mason et al., 2018), it is vital that we enhance our understanding of cognitive flexibility in autistic people, across the lifespan.

A clear profile of cognitive flexibility remains elusive, due to inconsistency across studies. Narrative reviews highlight the disparity of cognitive flexibility difficulties and methodological heterogeneity affecting outcomes and the interpretation of findings (Geurts et al., 2009; Hill, 2004; Russo et al., 2007). The most consistent pattern of results has been obtained with the Wisconsin Card Sorting Test (WCST) (Berg, 1948), with higher perseveration, i.e., the tendency to become stuck in set and persist with the same sorting strategy despite negative feedback, thought to specifically reflect cognitive flexibility difficulties in autistic people (Landry & Al-Taie, 2016). A meta-analysis of EF among autistic children and adolescents, identified cognitive flexibility as one of the core difficulties, with a moderate effect size that remained significant even after co-occurring ADHD and IQ were controlled for (Lai

et al., 2017). Nevertheless, there was substantial variability of effect sizes across studies. Similarly, in autistic adults, despite EF difficulties across all subdomains, cognitive flexibility was predominantly affected (Xie et al., 2020). Furthermore, Demetriou et al. (2018) examined the profile of EF difficulties across the lifespan in autistic people and obtained an overall moderate effect size. The only quantitative review to date focused specifically on cognitive flexibility in autistic people (Leung & Zakzanis, 2014) reported extensive variation in the magnitude of cognitive flexibility difficulties. However, this prior work includes self and parent-report measures and a broad range of tasks (e.g., set shifting, task switching and inhibitory control tasks), which together call into question the conclusions that can be drawn regarding cognitive flexibility in autistic people.

Several factors should be considered in light of the heterogeneity across studies, including the influence of task and sample characteristics. The inconsistent operationalisation of cognitive flexibility and the myriad of different paradigms likely contribute to the observed variability. Also, specific task features such as administration format (Demetriou et al., 2019) and type of instructions (Van Eylen et al., 2015) could moderate performance. The maturation of cognitive flexibility follows an inverted U-shaped curve, with a sharp increase during childhood, reaching a peak in early adulthood and deteriorating later in life (Zelazo et al., 2004). However, in autistic people, evidence suggests that this pattern is more complex, with higher interindividual variability found throughout development (Van Eylen et al., 2011). The prefrontal cortex (PFC) plays an important role in the development of cognitive flexibility (Buttelmann & Karbach, 2017). For instance, in a near-infrared spectroscopy study, during a set shifting task, neurotypical adults exhibited significant activation in the inferior PFC bilaterally and a similar pattern was observed in 5-year-

old children (Moriguchi & Hiraki, 2009). However, 3-year-olds who perseverated did not demonstrate significant activation in the right or left inferior PFC throughout the task. Some evidence suggests that autistic adolescents perform better than autistic children in cognitive flexibility tasks (D'Cruz et al., 2013; Van Eylen et al., 2015), and Lai et al. (2017) found a decrease in effect size by 0.062 for each year of increase in mean age. It is possible that maturational differences in autistic people, due to a protracted development, could explain some of the variability within and across studies and highlight the importance of examining cognitive flexibility within a developmental framework.

In summary, our understanding of cognitive flexibility in autistic people remains limited. Here, we diverge from previous meta-analyses broadly examining executive functioning, by honing in on fundamental forms of cognitive flexibility, thus addressing a notable gap in the literature. This focus gains added significance given the growing evidence that cognitive flexibility plays a key role across several domains, including educational attainment, mental health, and overall quality of life. Furthermore, we will also explore and quantify sources of heterogeneity, with an assessment of how tasks, outcomes, and age act as moderating factors. Taken together, this meta-analysis offers a more rigorous and in-depth understanding of cognitive flexibility across the lifespan in autistic people.

2. Methods

2.1. Eligibility Criteria

This systematic review and meta-analysis followed the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines (Moher et al., 2009). The inclusion criteria were defined a priori using the PICOS components below, as recommended by the PRISMA statement.

2.1.1. Participants

Participants with a diagnosis of ASD based on the Diagnostic and Statistical Manual of Mental Disorders (DSM; American Psychiatric Association, 2013) or the International Classification of Diseases (ICD; World Health Organization, 2019) criteria and/or a diagnosis using other valid diagnostic instruments, such as the Autism Diagnostic Observation Schedule (ADOS; Lord et al., 2012) and the Autism Diagnostic Interview (ADI; Rutter et al., 2003) were included. Subjects with learning disabilities (IQ < 70), related medical conditions (e.g., fragile X syndrome), and neurological disorders (e.g., epilepsy) were excluded. No restrictions regarding age were applied.

2.1.2. Interventions/Outcomes

Examining cognitive flexibility can be challenging due to its inconsistent operationalisation across the literature. Here, to address this conceptual ambiguity, we endeavour to more precisely define cognitive flexibility using set shifting paradigms (see below for details of specific tasks). While higher-level cognitive flexibility is often assessed with task switching paradigms, set shifting is lower down in the hierarchy of cognitive flexibility, serving as a scaffold for this vital cognitive ability

(Bunge & Zelazo, 2006), so there is a strong precedent for distinguishing between these two facets of cognitive flexibility and focusing on the most fundamental form. Furthermore, considering the heterogeneity across studies, to include other types of paradigms would introduce another source of heterogeneity for the quantitative synthesis of the evidence that could obscure vital distinctions at an outcome level and generate misleading results (Cooper, 2017). Additionally, set shifting paradigms are the most frequently used across the literature, with perseveration, thought to specifically reflect difficulties in cognitive flexibility (Hill, 2004; Landry & Al-Taie, 2016).

2.1.3. Comparators

Participants with no reported history of ASD, neurological or psychiatric conditions, without learning disabilities (IQ ≥ 70), matched on at least one IQ measure (e.g., nonverbal IQ) were included as the comparison group.

2.1.4. Study Design

Both cross-sectional and longitudinal designs were included, and for the latter only the baseline data was considered, to avoid practice effects. No other restrictions were applied.

2.2. Literature Search

Four separate electronic searches were performed using Pubmed, Embase, PsycInfo and Scopus as databases and a combination of the following terms: ‘autism’ and variations thereof, ‘Asperger’, ‘pervasive developmental disorder’ and variations thereof, ‘cognitive flexibility’ and variations thereof, ‘executive functioning’ and variations thereof, ‘set shift’ and several paradigms. The full search strategy is

available in Appendix A. The searches were limited to studies in humans, published in English between 1980 (first inclusion of autism diagnosis in the DSM-III) and November 2022. Manual searches were performed independently by two authors (CL and ESS) using the reference lists of included studies and previous systematic reviews.

2.3. Study Selection

After de-duplication, all records obtained from the electronic searches were sequentially screened by title and abstract. Subsequently, the full text of the remaining articles was examined independently by two authors (CL and ESS) using a piloted eligibility criteria checklist of the PICOS components.

2.4. Data Extraction

The following data was extracted from each included study: age range, mean age and standard deviation, task, sample size, number of males and females, diagnostic criteria and tool, IQ measurement tool, matching criteria (sex, age, IQ), and mean full-scale, performance/nonverbal and verbal IQs, where applicable. Outcomes were extracted as means and standard deviations for the autistic and comparison group at a single time point. Studies with more than one experiment with no overlap in participants were included separately. **Table 1** has a summary of all included studies.

2.5. Set shifting Paradigms and Task Outcome Measures

2.5.1. Intra-Extra Dimensional Set Shift (IED, Robbins et al., 1994)

The IED from the Cambridge Neuropsychological Test Automated Battery is a test involving rule acquisition and reversal with two dimensions, i.e., shape and colour (pink shapes and white lines). In the first stages of the task, the shifts are intra-dimensional, and participants must learn through trial and error based on feedback

which pink shape is the correct one, and following six consecutive correct responses, the subsequent stage begins with a different shape becoming the new rewarded stimulus. At stage eight the extra-dimensional shift occurs, i.e., lines become the new correct dimension, and at the final subsequent stage, the previously non-rewarded line becomes the correct stimulus. If participants do not reach the criterion of six consecutive correct responses by the fiftieth trial of each stage, then the IED is discontinued (Downes et al., 1989). The most frequently reported outcome was extra-dimensional shift errors, i.e., failure to shift dimension when the white lines become the correct dimension.

2.5.2. Wisconsin Card Sorting Test (WCST, Berg, 1948)

In the traditional WCST, participants must sort a total of 128 cards according to three categories, i.e., colour, shape, and number. The sorting rule must be inferred based on feedback and after ten consecutive correct trials the rule changes without warning. The most frequently reported outcome was perseverative errors, i.e., continuing to choose the previously reinforced category, despite negative feedback. Variations of the WCST, include reduced number of total cards, two sorting categories instead of three and different number of correct trials until the rule changes. Details of the variations included in each respective study are available in **Table 1**.

2.5.3. Modified Card Sorting Test (Nelson, 1976)

Participants must sort a total of 48 cards according to three dimensions, i.e., shape, number, and colour. Although similar to the WCST, the reduced 48-card deck includes no ambiguous cards (i.e., sharing more than one sorting dimension with the stimulus card). The correct sorting rule must be inferred based on feedback and after six consecutive correct trials, participants are explicitly instructed that the rule has

changed. Perseverative errors are defined as persisting to sort cards according to the previously correct dimension, following a rule switch.

2.5.4. Two-Choice Reversal Learning Task (D'Cruz et al., 2011)

Participants are shown two identical stimuli and must choose which one is in the correct location. Feedback is given indicating whether each choice was right or wrong. Without warning, a reversal of the correct location occurs after four, five or six consecutive correct trials. Perseverative errors are classified as errors following a reversal, when participants choose the previously rewarded choice before selecting the new correct response.

2.5.5. Probabilistic Reversal Learning Task (Cools et al., 2002)

Participants are shown two stimuli and must learn by trial and error which one is correct. The feedback given is probabilistic, i.e., participants are given inaccurate feedback on some trials. Reversal of the stimulus-reward contingency occurs without warning. The most frequently reported outcome was perseverative errors defined as errors that occur after a reversal in which participants continue to choose the previously reinforced response despite negative feedback. Details of the probabilistic reversal learning task variations (e.g., number of trials before reversal) included in each study are available in **Table 1**.

2.5.6. The Penn Conditional Exclusion Test (Kurtz et al., 2004)

This test involves four stimuli, in which three are matched based on either shape, size or line thickness. Participants must infer the sorting rule based on feedback and select the stimulus that does not belong. After ten consecutive correct trials, the sorting rule changes without warning. Regressive errors were defined as responses in

which participants after selecting the new correct response first, then reverted to sorting according to the previously reinforced rule, therefore indicating a difficulty sustaining a new response pattern, and instead reverting to the previously rewarded one.

2.5.7. Set Shifting Task (Hughes, 1998)

In this task, children must work out which cards are the teddy's favourite and sort them according to one of three dimensions, i.e., colour, shape, or size. The sorting rule must be inferred based on feedback and following six consecutive correct trials or after a maximum of 20 trials, the rule changes with the presentation of a new deck of cards and a different teddy. The reported outcome measure was total number of trials to criterion on the three sorting rules, i.e., a low number of trials needed to determine the sorting rule suggests greater cognitive flexibility (Pellicano et al., 2006).

2.5.8. Card Sorting Task (Velazquez et al., 2009)

In this task, participants must match different target stimuli with reference stimuli according to either colour or shape. Participants are given cues to indicate if the matching rules must be repeated or changed. Perseverative errors occurred when participants continued to match according to the previous rule.

2.5.9. Computerised Sequencing Game (Sawaya et al., 2019)

Eight stimuli are presented that differ on colour and shape in each trial (e.g., red and green, squares and stars). Participants must identify the correct sequence and the rules must be inferred based on feedback. In each trial there is a reversal halfway through, i.e., the first four stimuli are sequenced based on one rule and the subsequent

four based on another. Rule shift errors occurred after the reversal when participants failed to sequence according to the new rule.

2.6. Data Analysis

All analyses were performed in R (version 4.1.2) and the packages 'meta' and 'metafor' were used. The standardised mean difference between the autistic and comparison group was calculated as Hedges' g to correct for small sample bias (Hedges, 1980). Effect sizes were pooled together using a random-effects model (DerSimonian-Laird estimator for Tau^2) due to heterogeneity between studies. A random-effects model takes into account sampling error and between-study variance (Tau^2) when assigning weights, thus assuming a distribution of effect sizes (Cooper, 2017). A positive Hedges' g indicates that the comparison group performed better than the autistic group and the same effect size convention as with Cohen's d is applied, namely $g \approx 0.20$ is small, $g \approx 0.50$ is medium, $g \approx 0.80$ is large (Cohen, 1988). Heterogeneity was assessed using the Q-statistic, with a p-value of ≤ 0.10 indicating significant heterogeneity, i.e., variance in effect sizes is not due to sampling error alone (Higgins et al., 2019). The Q-statistic has low power in meta-analyses with either a small number of included studies or included studies with small sample sizes, thus a non-significant result should not be interpreted as proof of no heterogeneity. By convention a more stringent p-value of ≤ 0.10 is usually applied (Higgins et al., 2019).

Additionally, the I^2 statistic gives the percentage of total variance in effect sizes that is due to between-study heterogeneity (rather than sampling error), interpreted as follows: $I^2 = 25\%$ is low, $I^2 = 50\%$ is moderate, and $I^2 = 75\%$ is high (Higgins & Thompson, 2002). Heterogeneity was explored and subgroup analyses of task, outcome, and age were performed. For the latter, studies were categorised based on

mean age reported, consistent with other meta-analyses (e.g., Demetriou et al., 2018): 'children ≤ 12', 'adolescents >12 <18', and 'adults ≥ 18'. A meta-regression with mean age as a moderator was also performed. Here we followed the Cochrane guidance (Higgins et al., 2019), recommending that subgroup analyses and meta-regressions should have at least ten studies for each characteristic included. Although this is a general rule of thumb, we endeavoured to adhere to it as much as possible.

2.6.1. Publication Bias

Publication bias was assessed with a funnel plot visually inspected for asymmetry and formally evaluated with Egger's test (Egger et al., 1997). Studies with small samples require larger effect sizes in order to achieve significant results and thus are more likely to go unpublished (Borenstein et al., 2009). Funnel plots show the association between each study's effect size and standard errors, i.e., studies with larger samples have smaller standard errors. In case of no or minimal bias, studies will be spread evenly on each side of the pooled effect size, forming an inverted funnel shape. If there is publication bias, the funnel plot will be asymmetrical. The Egger's test quantifies the asymmetry with a p-value of ≤ 0.05 showing significant asymmetry. However, recent evidence suggests that assessing funnel plot asymmetry of standardised mean differences using the Egger's test may lead to an increase in false positive results (type I errors), especially when studies have small sample sizes or when there is high between-study heterogeneity (Pustejovsky & Rodgers, 2019). This occurs because there is an artifactual correlation between the standardised mean difference and its standard error, and therefore a modified test referred to as the Pustejovsky-Rodgers test (Pustejovsky & Rodgers, 2019) has been proposed and will also be reported.

2.6.2. Quality Assessment

The Newcastle-Ottawa Scale for non-randomised case-control studies (Wells et al., 2000) was used to assess the quality of each study. This scale contains a total of eight items covering selection, comparability, and exposure. The selection component considers whether case definition was adequate (i.e., with independent validation), the representativeness of cases, and the selection and definition of controls (i.e., in this case explicitly with no history of ASD and other conditions as per eligibility criteria). The comparability section encompasses whether cases and controls were matched and/or confounders adjusted for in the analysis, and the exposure section covers the ascertainment method and whether the same one was used for cases and controls, and finally the non-response rate. A star rating system is used, with the highest quality studies receiving one star per item and two stars in the comparability category, and up to a total of nine stars.

2.6.3. Influence Analysis

An influence analysis was performed using the leave-one-out method, i.e., the pooled effect size and heterogeneity (I^2) were recalculated omitting one study at a time, thus enabling the detection of studies that may exert a disproportionate influence and distort the overall results (Viechtbauer & Cheung, 2010).

3. Results

3.1. Study Selection and Characteristics

A total of 7989 records were identified through database searching and an additional 13 through manual searching. The PRISMA flow diagram presented in **Figure 1** provides details of the study selection process. Fifty-nine studies were included in the quantitative synthesis and comprised of 4158 participants in total, 2122 in the autistic group, and 2036 in the comparison group. Ages ranged from 4 to 85 years, with a mean age of 17.8 for the autistic group and 18 for the comparison group. The search was repeated in August 2023, and this did not yield any additional studies for inclusion.

3.2. Primary Meta-analysis

The pooled standardised mean difference between the autistic and comparison group was small to moderate and statistically significant (g = 0.44, 95% CI 0.33-0.55, $p < 0.001$), i.e., the autistic group had significantly more difficulties in cognitive flexibility. However, there was significant moderate heterogeneity between studies ($I^2 = 64\%$, $p < 0.001$). As shown by the forest plot in **Figure 2**, there is substantial variance in effect sizes, ranging from g = -0.99 to g = 2.22.

3.3. Subgroup Analyses

3.3.1. Effect of Task

Only the IED and WCST (and variations thereof) had sufficient numbers of studies to include in the subgroup analysis (Higgins et al., 2019), thus fourteen studies were excluded. There was no significant subgroup difference between tasks ($p = 0.18$),

however the WCST had the largest effect size (g = 0.58, 95% CI 0.37-0.79). Results are summarised in **Table 2.**

3.3.2. Effect of Outcome Measure

Only perseverative errors and extra-dimensional shift errors had sufficient numbers of studies to include in the subgroup analysis (Higgins et al., 2019), thus seven studies were excluded. A significant subgroup difference between outcomes (p = 0.05) was found, with perseverative errors showing the largest effect size (g = 0.48, 95% CI 0.35-0.62). Results are summarised in **Table 3**.

3.3.3. Effect of Age

No significant difference was found between subgroups of children (≤ 12), adolescents (>12 <18) and adults (≥ 18) (p = 0.27). The adult subgroup had the largest effect size (g = 0.54, 95% CI 0.34-0.73). Results are summarised in **Table 4**.

3.4. Meta-regression

Age was not a significant moderator (B = 0.005, 95% CI -0.01-0.02, p = 0.47) and did not explain any of the variance in effect sizes (R^2 = 0%). One study was not included in this analysis due to a significant age difference between the autistic and comparison group (Geurts et al., 2020). A meta-regression plot is presented in **Figure 3**.

3.5. Publication Bias

The funnel plot is presented in **Figure 4.** The majority of studies are scattered relatively evenly around the pooled effect size, although some asymmetry can be observed. The Egger's test (p = 0.04) suggests that there could be some publication bias, however the Pustejovsky-Rodgers test was not significant (p = 0.12). It can be

assumed that there is minimal publication bias and that the pooled effect size is fairly representative.

3.6. Quality Assessment

The majority of studies were rated as having overall adequate quality ($M = 6.86$, $SD = 1.54$). Results are summarised in **Table 5**.

3.7. Influence Analysis

The influence analysis showed that for both pooled effect size and heterogeneity, there is minimal impact of any individual study on the reported findings. Results are shown in **Figures 5 and 6**.

4. Discussion

The present meta-analysis showed the existence of cognitive flexibility difficulties in autistic people, in the absence of learning disabilities. However, there was also significant moderate heterogeneity between studies, and it is possible that this variance could at least be partially explained by a broad profile of cognitive flexibility difficulties in autistic people. Subgroup analyses revealed a significant difference between task outcomes, with perseverative errors obtaining the largest effect size, consistent with previous results (e.g., Landry & Al-Taie, 2016), yet it is important to acknowledge that power might be an issue here since forty-three included studies had perseverative errors as an outcome measure, whereas only nine reported extra-dimensional shift errors. Although a large body of research has demonstrated that perseveration constitutes a difficulty in autistic people, it would be misleading to conclude that cognitive flexibility difficulties are due to perseveration alone, as more studies are needed to compare the sensitivity of other outcome measures. For instance, in D'Cruz et al. (2013), there was no significant difference in perseverative errors, however the autistic group made significantly more regressive errors, which were positively correlated with behavioural rigidity, indexed by clinical ratings of RRBs. Moving forward, it is also important to consider a more fine-grained assessment of perseveration, for instance in probabilistic reversal learning tasks, by distinguishing between perseveration following a reversal and perseveration following negative probabilistic feedback. The latter case might indicate that participants are more readily inferring contextual changes in the environment, which has been associated with both autism and anxiety (Browning et al., 2015; Lawson et al., 2017).

Additionally, in the subgroup analyses, no significant differences between tasks were found, nonetheless, the WCST obtained the largest effect size. The specificity of the WCST is often debated (e.g., Nyhus & Barcelo, 2009) as task performance relies on multiple EF domains, including working memory and inhibition (Russo et al., 2007). This multifactorial EF demand has been highlighted in neuroimaging research, demonstrating fractionation of cognitive components that are integrated to perform the task (Buchsbaum et al., 2005). Notwithstanding the criticism, as evidence suggests that EF is divided into separable, yet correlated component processes (Miyake & Friedman, 2012), it may not be feasible to develop a completely 'pure' cognitive flexibility task. Furthermore, Van Eylen et al. (2015), found that even after working memory and inhibition difficulties were controlled for, autistic people still showed significantly more difficulties in cognitive flexibility, with higher perseveration.

As in other meta-analyses (e.g., Lai et al., 2017), the IED task obtained the smallest effect size. It has been suggested that the differences captured by the IED could be partially due to difficulties in sustaining attention, as cognitive flexibility is only assessed at the end of the task when the extra-dimensional shift occurs (Geurts et al., 2009). Consistently, Sinzig et al. (2008), compared the performance of autistic children with and without co-occurring ADHD and found that only the former showed difficulties in the task. In contrast with other tasks such as the WCST that assess cognitive flexibility throughout, it is possible that the IED's stepwise design with the extra-dimensional shift at the end, is not able to fully capture the extent of cognitive difficulties in autistic people. However, it is also important to consider that the IED task terminates after 50 trials on any stage, if the learning criterion of six consecutive correct responses is not achieved (Downes et al., 1989). In this sense, it is possible that participants with greater EF difficulties do not reach stage eight, where the extra-

dimensional shift occurs. An adjustment is therefore required, in which 25 errors should be added for each missed stage of the task (Geurts et al., 2009). If studies failed to adjust errors in this manner, it would lead to differences between participants being obscured.

The measurement of cognitive flexibility is challenging due to its inherent complexity, intricate relationships with other EF domains and changeable nature throughout the lifespan. In addition to the myriad of different paradigms, specific administration factors, such as type of instructions given can affect cognitive flexibility (Van Eylen et al., 2011), however this level of detail is rarely reported in studies. Furthermore, the ecological validity of neuropsychological tasks is often debated, as they do not always converge with self-report measures (Toplak et al., 2013). Despite the apparent face-validity, there is a discrepancy between cognitive flexibility difficulties and the prominent 'real-world' behavioural flexibility challenges, referred to as the 'paradox of cognitive flexibility in autism' (Geurts et al., 2009). Moving forward, more research is needed to address this 'paradox', using cognitive tasks that confer greater ecological validity in combination with self-report measures to enable a more comprehensive investigation of flexibility in autistic people. These new tools would be best co-designed with the autism community in order to have the greatest impact.

Reversal learning tasks are a widely used translational paradigm to index flexibility (Uddin, 2021) and may offer a better trade-off between construct and ecological validity. For instance, in probabilistic reversal learning (D'Cruz et al., 2013; Weiss et al., 2021) the uncertainty in the task is captured by the probabilistic reinforcement schedules, and thus the ability to learn about this uncertainty and

flexibly respond to variable contingencies, more closely resembles the 'real-world' flexibility demands of continuously changing environments. Additionally, reversal learning task responses are amenable to computational modelling, which can reveal the latent mechanisms that drive behavioural differences, however there is a paucity of studies adopting computational modelling in autistic people at present (though see Crawley et al., 2020; Lawson et al., 2017; Manning et al., 2017 for notable exceptions). Furthermore, cognitive flexibility difficulties in probabilistic reversal learning tasks have been associated with RRBs in autistic people (Crawley et al., 2020; D'Cruz et al., 2013). One possibility is that measured difficulties in probabilistic reversal learning tasks could be due to higher response monitoring requirements, i.e., the ability to evaluate behavioural consequences and adjust accordingly to optimise outcomes (Thakkar et al., 2008). It has been suggested that in autistic people, structural and functional alterations of the anterior cingulate cortex might underlie response monitoring difficulties and thus contribute to behavioural rigidity (Thakkar et al., 2008). Future neuroimaging studies should explore the role of the anterior cingulate cortex during probabilistic reversal learning tasks and the link with RRBs among autistic individuals.

The meta-regression showed that age did not account for any of the variance between studies and in the subgroup analysis there was no significant difference between the three age groups. Assessing age effects in this meta-analysis was hampered by the fact that many studies included very wide age ranges in their samples. Nevertheless, we note that the lowest effect size for cognitive flexibility difficulties was found in the adolescent group and the highest in the adult group. Adolescence is a developmental period of substantial neural changes, including synaptic reorganisation, that could contribute to the reduced differences between autistic and

typically developing adolescents (Blakemore & Choudhury, 2006). Additionally, it has been suggested that in autistic adults age-related cognitive decline disproportionately affects some domains including cognitive flexibility, yet with substantial interindividual variability (Powell et al., 2017). Few studies have focused on cognitive functioning among older autistic adults, with some notable exceptions (e.g., Geurts et al., 2020). Moving forward, longitudinal or accelerated-longitudinal studies are sorely needed to enable a comprehensive understanding of cognitive flexibility across the lifespan.

4.1. Limitations and Future Directions

One important caveat to consider is that several studies had mixed samples with wide age ranges, some spanning across childhood and into adulthood (e.g., 6 to 44 years - Miller et al., 2015). To perform the subgroup analysis and meta-regression we had to rely on the study means and it is therefore likely that this led to age-related differences in cognitive flexibility being obscured. Future empirical studies of cognitive flexibility in autistic people should endeavour to employ a stratified or longitudinal approach, to enable more precise estimates of cognitive flexibility difficulties across the lifespan. This would help to identify developmental periods when autistic people might require more support or adjustments to cope with the challenging and changing demands of real-life.

Although participants were matched on at least one IQ measure (e.g., verbal IQ or performance IQ), matching criteria differed substantially across studies, thus not permitting subgroup analyses to be performed, however it is possible that this contributed to the heterogeneity observed. Finding appropriate matching strategies can be a challenge due to the distinctive profiles of cognitive strengths and difficulties in

autistic people, and possible limitations of standard assessment tools to capture these. It has been proposed that one possible avenue is to match on an area of functioning upon which the task heavily relies on, such as verbal abilities, thus allowing for differences to be controlled for (Burack et al., 2004). In the present study we carefully excluded for the presence of learning disabilities as this would confound the profile of cognitive flexibility difficulties, however more research is needed to explore EF across a wide range of abilities within the autistic spectrum, to enable greater generalisability of findings.

The majority of studies did not report the severity of ASD symptoms (e.g., ADOS scores) and this might represent another source of heterogeneity in the observed effects across studies. Cognitive flexibility difficulties have been linked with more pronounced RRBs (e.g., Lopez et al., 2005), therefore it is important for future studies with complete outcome reporting to explore the association between autistic symptom profiles and cognitive flexibility. Furthermore, due to lack of reporting across studies, the impact of co-occurring conditions could not be considered (though see Braden et al., 2017; Crawley et al., 2020; Kilincaslan et al., 2010; Van Eylen et al., 2015 for notable exceptions). Anxiety disorders are estimated to affect around 40 percent of autistic individuals (Hollocks et al., 2019; van Steensel et al., 2011). Given this high prevalence rate and that anxiety is known to have a deleterious effect on cognitive flexibility (Park & Moghaddam, 2017; Wilson et al., 2018), it is vital for future studies to take this overlap into consideration. ASD is also highly co-occurring with ADHD (Hofvander et al., 2009), however before the DSM-5 (American Psychiatric Association, 2013) these two diagnoses were mutually exclusive, thus limiting research studying these two conditions together. The EF difficulties in ADHD are well-documented (Craig et al., 2016; Happé et al., 2006; Sinzig et al., 2008) and

growing evidence suggests that autistic individuals with co-occurring ADHD have more pronounced cognitive difficulties (Craig et al., 2016; Dajani et al., 2016), thus emphasising the need to take co-occurring conditions into account when considering cognitive flexibility in autistic people.

In summary, this meta-analysis has highlighted the existence of cognitive flexibility difficulties among autistic individuals in the absence of intellectual disabilities, but also that this profile is characterised by extensive heterogeneity. While several potential contributing factors have been discussed in the preceding sections, it is essential to consider the possibility that this heterogeneity could represent the broad range of cognitive flexibility profiles across the autistic spectrum. The present findings have therefore important ramifications for support and adjustments for autistic people. Considering the prevalence of cognitive flexibility difficulties throughout the lifespan in autistic people, links with poorer outcomes (e.g., Bertollo et al., 2020) and the known plasticity of cognitive flexibility particularly during childhood (Buttelmann & Karbach, 2017), very little research has focused on interventions that might support better cognitive flexibility abilities, or strategies to manage everyday situations that require substantial cognitive flexibility burden. Evidence to date from randomised controlled trials on cognitive remediation strategies remains inconsistent (Pugliese et al., 2020), with cognitive enhancement therapy among autistic adults (Eack et al., 2018) and a cognitive behavioural intervention designed for autistic children (Kenworthy et al., 2014) showing some promising results. In this context it is worth noting that RRBs, and the perseveration characteristic of cognitive inflexibility, may function as coping mechanisms for autistic people in situations where they feel overwhelmed, uncertain, or anxious (Collis et al., 2022; Joyce et al., 2017), and the

development of new interventions to support cognitive flexibility should consider how to optimise the positive and minimise the negative functions of these behaviours.

Strategies to adjust environments to accommodate the needs of autistic people in terms of cognitive flexibility should also be explored. It is crucial to prioritise adaptations or adjustments to the environment wherever possible so that the burden does not disproportionately lie with autistic people to change or adapt themselves. Simple accommodations to work or education settings might include cueing or signalling upcoming changes to tasks in advance, to reduce cognitive flexibility demands. Nonetheless, it is also important to acknowledge that everyday life is inherently filled with unpredictability. For example, an unexpected burst pipe might cause roadworks that disrupt your morning commute and require a flexible adjustment to your usual route. To alleviate distress in these sorts of circumstances some autistic people may choose to engage with support aimed at bolstering cognitive flexibility. This may be especially useful to reduce anxiety, which is elevated in autistic individuals (Hollocks et al., 2019; van Steensel et al., 2011), has been persistently linked to a dislike or intolerance of uncertainty (Jenkinson et al., 2020), and represents a major priority of the autism community (Autistica, 2016). Prioritising mental health support is imperative, and continued research into the heterogeneity of cognitive flexibility across the autism spectrum may yield valuable insights to inform personal decisions regarding individualised coping strategies, interventions, and new avenues for support.

In this context, it is also vital to consider various cognitive strengths and exceptional abilities among autistic people such as, enhanced visual-spatial processing (Falter et al., 2008; O'Riordan et al., 2001) and highly focused attention (Murray et

al., 2005). Cognitive theories of autism provide different perspectives on the intertwined profile of cognitive strengths and difficulties in autistic people. For instance, an update to the Weak Central Coherence account (Happé & Frith, 2006) posits that autistic individuals have a detail-focused cognitive style with superior local processing. However, a detail-focused processing style may also pose challenges in integrating multiple sources of information into a coherent whole and understanding a broader context, critical for the flexible adjustment of cognitive strategies in response to varying contextual demands. Another prominent theory, Monotropism (Murray et al., 2005), proposes that autistic people have a cognitive style characterised by a narrower and more intense focus of attention. Here, the mind is proposed as an interest-based system that guides the allocation of finite attentional resources, which are in competition among different cognitive processes (Garau et al., 2023). This enhanced focus could be highly advantageous in cognitive tasks requiring great attention to detail, yet present a challenge where rapid shifting of attention between different stimuli is required. This may manifest as the reported cognitive flexibility difficulties in this meta-analysis. Additionally, Bayesian accounts of autism, suggest that autistic people tend to place greater weight on new incoming sensory inputs relative to prior expectations (e.g., Pellicano & Burr., 2012). This may result in an enhanced ability to detect novel changes in the environment (Stark et al., 2021), but also contribute to an overestimation of uncertainty (Lawson et al., 2017). In this context, striving for sameness or seemingly 'inflexible' cognitive strategies could be construed as adaptive and highly appropriate responses. The full complexity of the cognitive landscape across the autism spectrum highlights the need for a nuance in the development of theoretical perspectives that consider both the challenges and the unique cognitive assets possessed by autistic individuals.

Moving forward, it is vital to explore the heterogeneity within the autistic spectrum and investigate whether there are subgroups with more homogenous cognitive flexibility profiles, to enable the progression from a 'one-size-fits-all' approach (Lombardo et al., 2019) towards the development of targeted, autism-specific support tools that consider individual profiles of strengths and difficulties. The emerging field of computational psychiatry, particularly unsupervised machine learning techniques, allow the discovery of hidden structures in data, without the assumption of prior knowledge or labels, and have therefore been used in the identification of previously undetected subtypes within the autistic spectrum (Stevens et al., 2019; Zheng et al., 2020). Furthermore, these data-driven approaches could be combined with the parameters from theory-driven models of behaviour, such as in probabilistic reversal learning tasks, to improve the precision of identified subtypes in service of designing better and more mechanistically grounded interventions for autistic people. The present findings should pave the way for research to parse out the heterogeneity of cognitive flexibility, and ultimately improve individual support and outcomes for autistic people.

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Declaration of Competing Interest

The authors report no declarations of interest.

Appendix A Search Strategy

Database: Embase

1 autis*.mp.
2 "autism spectrum disorder*".mp.
3 "ASD".mp.
4 Asperger*.mp.
5 "pervasive developmental disorder not otherwise specified".mp.
6 "pervasive developmental disorder*".mp.
7 1 or 2 or 3 or 4 or 5 or 6
8 "cognitive flexibility".mp.
9 "mental flexibility".mp.
10 "cognitive rigidity".mp.
11 "mental rigidity".mp.
12 "cognitive inflexibility".mp.
13 "mental inflexibility".mp.
14 "executive function*".mp.
15 "executive dysfunction*".mp.
16 "set shift*".mp.
17 "Intra-Extra Dimensional Set Shift*".mp.
18 "Dimensional Change Card Sort*".mp.
19 "Flexible Item Selection".mp.
20 "Modified Card Sort*".mp.
21 "Wisconsin Card Sort*".mp.
22 "reversal learning".mp.
23 "probabilistic learning".mp.
24 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23
25 7 and 24
26 limit 25 to (human and english language and yr="1980 -Current")

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Figure Captions

Figure 1. PRISMA Flow Diagram of Study Selection

Figure 2. Forest Plot of Included Studies

The diamond shape represents the pooled standardised mean difference (g = 0.44) between the autistic and comparison group. Positive Hedges' g indicates that the comparison group had lower cognitive flexibility difficulties. Square sizes vary according to the weight attributed to each study under the random-effects model.

Figure 3. Meta-regression Bubble Plot with Age as a Moderator

Each bubble shows an individual study, and the size varies according to the weight assigned under the random-effects models, i.e., studies with larger samples are assigned higher weights as displayed by the larger bubbles. The green line represents the regression line of best fit. Although the standardised mean difference appears to increase slightly with age, the dispersion shows that age is not a significant moderator.

Figure 4. Funnel Plot of Included Studies

Each dot shows an individual study (studies with larger samples have smaller standard errors). The vertical line represents the pooled effect size (g = 0.44) and the diagonal lines (funnel) represent the 95% confidence interval. Although most studies are spread relatively evenly around the pooled effect size, i.e., there appears to be minimal publication bias, some asymmetry can be observed.

Figure 5. Forest Plot of Influence Analysis - Sorted by Effect Size (low to high)

This forest plot shows the results of the influence analysis using the leave-one-out method. On the right-hand side is the recalculated pooled effect size, 95% confidence interval, and heterogeneity (I^2), with the corresponding omitted study on the left-hand

side. The pooled effect size and 95% confidence interval (g = 0.44, 95% CI 0.33-0.55) of the original meta-analysis are represented by the dashed line and green area, respectively. As displayed, the overall results were not disproportionally influenced by any individual study, as the recalculated pooled effect sizes range from 0.42 to 0.46.

Figure 6. Forest Plot of Influence Analysis - Sorted by Heterogeneity (low to high)

This forest plot shows the results of the influence analysis using the leave-one-out method. On the right-hand side is the recalculated pooled effect size, 95% confidence interval, and heterogeneity (I^2), with the corresponding omitted study on the left-hand side. The pooled effect size and 95% confidence interval (g = 0.44, 95% CI 0.33-0.55) of the original meta-analysis are represented by the dashed line and green area, respectively. As displayed, the overall results were not disproportionally influenced by any individual study, given that the heterogeneity (I^2) of the original meta-analysis is 64 percent and the recalculated heterogeneity (I^2) ranges from 61 to 65 percent.